SG No. 201 T_{h}^2 $Pn\bar{3}$ [cubic]

* plus set: $+[0,\ 0,\ 0]$

Table 1: Wyckoff site: 2a, site symmetry: 23.

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12]
2	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	[13,14,15,16,17,18,19,20,21,22,23,24]

Table 2: Wyckoff site: 4b, site symmetry: .-3.

No.	position	mapping
1	$[0,\ 0,\ 0]$	[1,5,9,13,17,21]
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[2,7,12,14,19,24]
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[3,8,10,15,20,22]
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	[4,6,11,16,18,23]

Table 3: Wyckoff site: 4c, site symmetry: .-3.

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[1,5,9,13,17,21]
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	[2,7,12,14,19,24]
3	$\left[0,\ \frac{1}{2},\ 0\right]$	[3,8,10,15,20,22]
4	$\left[\frac{1}{2},\ 0,\ 0\right]$	[4,6,11,16,18,23]

Table 4: Wyckoff site: 6d, site symmetry: 222..

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	[1,2,3,4]
2	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	[5,6,7,8]
3	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	[9,10,11,12]
4	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	[13,14,15,16]
5	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	[17,18,19,20]
6	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	[21,22,23,24]

Table 5: Wyckoff site: 8e, site symmetry: .3.

No.	position	mapping
1	$[x,\ x,\ x]$	[1,5,9]
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x\right]$	[2,7,12]
3	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}-x\right]$	[3,8,10]
4	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[4,6,11]
5	$[-x,\ -x,\ -x]$	[13,17,21]
6	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -x\right]$	[14,19,24]
7	$\left[x+\frac{1}{2},\ -x,\ x+\frac{1}{2}\right]$	[15,20,22]

continued ...

Table 5

No.	position	mapping
8	$\left[-x,\, x+\frac{1}{2},\, x+\frac{1}{2}\right]$	[16,18,23]

Table 6: Wyckoff site: 12f, site symmetry: 2..

No.	position	mapping
1	$\left[x,\, \frac{1}{4},\, \frac{1}{4}\right]$	[1,4]
2	$\left[\frac{1}{2}-x,\, \frac{1}{4},\, \frac{1}{4}\right]$	[2,3]
3	$\left[\frac{1}{4},\, x,\, \frac{1}{4}\right]$	[5,8]
4	$\left[\frac{1}{4},\, \frac{1}{2}-x,\, \frac{1}{4}\right]$	[6,7]
5	$\left[\frac{1}{4},\, \frac{1}{4},\, x\right]$	[9,12]
6	$\left[\frac{1}{4},\, \frac{1}{4},\, \frac{1}{2}-x\right]$	[10,11]
7	$\left[-x,\, \frac{3}{4},\, \frac{3}{4}\right]$	[13,16]
8	$\left[x+\frac{1}{2},\, \frac{3}{4},\, \frac{3}{4}\right]$	[14,15]
9	$\left[\frac{3}{4},\, -x,\, \frac{3}{4}\right]$	[17,20]
10	$\left[\frac{3}{4},\, x+\frac{1}{2},\, \frac{3}{4}\right]$	[18,19]
11	$\left[\frac{3}{4},\, \frac{3}{4},\, -x\right]$	[21,24]
12	$\left[\frac{3}{4},\, \frac{3}{4},\, x+\frac{1}{2}\right]$	[22,23]

Table 7: Wyckoff site: 12g, site symmetry: 2..

No.	position	mapping
1	$\left[x,\, \frac{3}{4},\, \frac{1}{4}\right]$	[1,4]
2	$\left[\frac{1}{2}-x,\, \frac{3}{4},\, \frac{1}{4}\right]$	[2,3]
3	$\left[\frac{1}{4},\, x,\, \frac{3}{4}\right]$	[5,8]
4	$\left[\frac{1}{4},\, \frac{1}{2}-x,\, \frac{3}{4}\right]$	[6,7]
5	$\left[\frac{3}{4},\, \frac{1}{4},\, x\right]$	[9,12]
6	$\left[\frac{3}{4},\, \frac{1}{4},\, \frac{1}{2}-x\right]$	[10,11]
7	$\left[-x,\, \frac{1}{4},\, \frac{3}{4}\right]$	[13,16]
8	$\left[x+\frac{1}{2},\, \frac{1}{4},\, \frac{3}{4}\right]$	[14,15]
9	$\left[\frac{3}{4},\, -x,\, \frac{1}{4}\right]$	[17,20]
10	$\left[\frac{3}{4},\, x+\frac{1}{2},\, \frac{1}{4}\right]$	[18,19]
11	$\left[\frac{1}{4},\, \frac{3}{4},\, -x\right]$	[21,24]
12	$\left[\frac{1}{4},\, \frac{3}{4},\, x+\frac{1}{2}\right]$	[22,23]

Table 8: Wyckoff site: 24h, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[2]
3	$\left[\frac{1}{2}-x,\, y,\, \frac{1}{2}-z\right]$	[3]
4	$\left[x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[4]
5	$[z,\, x,\, y]$	[5]
6	$\left[z,\, \frac{1}{2}-x,\, \frac{1}{2}-y\right]$	[6]

continued ...

Table 8

No.	position	mapping
7	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y\right]$	[7]
8	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-y\right]$	[8]
9	$[y,\ z,\ x]$	[9]
10	$\left[\frac{1}{2}-y,\ z,\ \frac{1}{2}-x\right]$	[10]
11	$\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[11]
12	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x\right]$	[12]
13	$[-x,\ -y,\ -z]$	[13]
14	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[14]
15	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	[15]
16	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16]
17	$[-z,\ -x,\ -y]$	[17]
18	$\left[-z,\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	[18]
19	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ -y\right]$	[19]
20	$\left[z+\frac{1}{2},\ -x,\ y+\frac{1}{2}\right]$	[20]
21	$[-y,\ -z,\ -x]$	[21]
22	$\left[y+\frac{1}{2},\ -z,\ x+\frac{1}{2}\right]$	[22]
23	$\left[-y,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[23]
24	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ -x\right]$	[24]